

Operational Alignment of Aurora-Lens with OECD Due Diligence Guidance for Responsible AI (2026)

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Abstract

The 2026 OECD *Due Diligence Guidance for Responsible AI* establishes a six-step Responsible Business Conduct (RBC) framework that applies across the AI system value chain. The guidance is explicit about what due diligence requires; it is silent about how runtime enforcement should be implemented.

This paper examines Aurora-Lens, a deterministic transport-layer governance proxy, and maps its mechanisms to each step of the OECD framework. The central argument is that Aurora-Lens converts procedural due diligence expectations into enforceable runtime controls — operating between applications and LLM providers without requiring any modification to underlying foundation models.

1. Introduction

LLM deployment in regulated environments has outpaced the governance infrastructure available to manage it. The mitigation approaches currently in common use — prompt engineering, output filtering, fine-tuning, retrieval augmentation, human review — each address surface-level risks. None of them establish structural guarantees about whether a system output is legitimate.

The OECD *Due Diligence Guidance for Responsible AI* (2026) reframes the governance question. Rather than asking what a model can do, it asks what enterprises are accountable for. The framework requires organisations to embed RBC into management systems, identify and mitigate adverse impacts, track implementation, communicate what has been done, and enable remediation when things go wrong.

What the guidance does not provide is a runtime enforcement architecture. It describes the shape of due diligence without specifying what it looks like in operation.

Aurora-Lens is designed to fill that gap. Positioned at the transport layer between an application and an LLM provider, it applies deterministic admissibility constraints to model outputs before they reach end users and records every governance decision in a tamper-evident audit ledger. This paper maps that architecture to the OECD six-step framework.

2. OECD Responsible AI Due Diligence Framework

The OECD guidance extends the MNE Guidelines' RBC due diligence model to AI systems, with six iterative steps:

1. Embed RBC into policies and management systems
2. Identify and assess actual and potential adverse impacts
3. Cease, prevent, and mitigate adverse impacts
4. Track implementation and results

5. Communicate how impacts are addressed
6. Provide for or cooperate in remediation

The framework is risk-based, iterative, and context-sensitive. It applies across the full AI lifecycle. Throughout, it emphasises transparency and traceability, stakeholder engagement, cross-jurisdictional coherence, and documentation of what mitigation actions were taken.

Notably, the framework does not prescribe technical architecture. It establishes governance expectations that must be demonstrable — and leaves organisations to determine how.

3. Aurora-Lens Governance Architecture

3.1 Architectural Placement

Aurora-Lens occupies the transport layer:

Application → Aurora-Lens Governance Proxy → LLM Provider → Tamper-Evident Audit Ledger

Governance is applied post-generation but pre-delivery. The underlying model is not modified.

3.2 Core Mechanisms

Aurora-Lens implements a deterministic governance architecture composed of the following interlocking mechanisms:

- **Admissibility Gating** — Interpretive commitments must satisfy explicit structural constraints before being delivered. Outputs are not evaluated solely for content appropriateness; they are evaluated for whether the conclusion is epistemically permitted given the available evidence structure.
- **Parallel Interpretation with Constraint-Forced Collapse** — Multiple admissible interpretations are maintained simultaneously. Collapse to a single determinate conclusion is permitted only when structural constraints eliminate all alternatives. Heuristic preference or statistical likelihood does not trigger resolution.
- **Deterministic Intervention Actions** — Governance outcomes are explicit and reproducible: PASS, SOFT_CORRECT, FORCE_REVISE, or HARD_STOP. These interventions occur pre-delivery and are governed by rule-based constraint evaluation rather than probabilistic confidence.
- **Refusal as First-Class Outcome** — Withholding an output is a legitimate governance decision when admissibility conditions are not met. Refusal is treated as structurally correct, not as system failure.
- **Persistent Existence Frame (PEF)** — The foundational substrate of Aurora-Lens. Deterministic due diligence enforcement requires invariant state across interactions. Conventional temporal modelling — whether session history, probabilistic recall, or heuristic memory — permits semantic drift and undermines reproducibility.

PEF replaces conventional temporal sequencing with a continuous present-state structure in which roles, constraints, interpretive commitments, and collapse conditions persist until structurally resolved. Past states are reconstructed from retained structural traces within the frame; future projections are generated from the same invariant substrate. There is no external timeline driving collapse — only constraint satisfaction.

This non-temporal persistence ensures that governance decisions are replayable and audit-stable. Given identical input and policy configuration, the system produces identical admissibility outcomes.

This property is essential for satisfying OECD requirements concerning traceability, documentation, and reproducible remediation.

- **Tamper-Evident Audit Ledger** — Every governance decision is recorded in a hash-chained, line-delimited JSON format (AFL-JSONL). Entries include decision state, constraint evaluation, and intervention outcome. Because governance decisions arise from deterministic state within PEF, audit reconstruction is mechanically verifiable rather than interpretively inferred.
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4. Alignment with OECD Due Diligence Steps

4.1 Embed RBC into Policies and Management Systems

The OECD framework requires governance to be embedded structurally — not applied as an advisory layer after the fact.

Aurora-Lens satisfies this requirement by treating governance as infrastructure. Policy is defined in version-controlled YAML. Enforcement occurs at the transport layer, before output delivery, and cannot be bypassed through prompt modification. Governance is not a recommendation; it is a constraint on what the system is permitted to deliver.

4.2 Identify and Assess Adverse Impacts

Risk identification in Aurora-Lens is formalised rather than heuristic. An explicit flag taxonomy covers categories including contradicted fact, hallucinated entity, and temporal inconsistency. Outputs are evaluated against persistent state rather than against the current request in isolation, and ambiguity is maintained until it can be structurally resolved.

This makes risk detection reproducible: the same input, against the same persistent state, produces the same assessment.

4.3 Cease, Prevent, and Mitigate Adverse Impacts

Mitigation in Aurora-Lens is deterministic and pre-delivery. `HARD_STOP` blocks unsafe outputs entirely. `FORCE_REVISION` allows a bounded corrective attempt before delivery. Confident error — a model asserting something structurally unjustifiable with high apparent certainty — triggers deterministic refusal.

The key property here is that mitigation happens before the output reaches a user. It is not a post-hoc review process.

4.4 Track Implementation and Results

Every governance decision is permanently recorded in a hash-chained forensic ledger. Entries carry unique content identifiers, and governance traces are session-isolated. Decision states are reproducible: it is possible to reconstruct exactly what state the system was in when a particular decision was made.

This provides audit-grade traceability, not merely logging.

4.5 Communicate How Impacts Are Addressed

Aurora-Lens produces structured decision objects that can be exported as audit logs and shared with stakeholders. Trace IDs (for observability) and audit IDs (for accountability) are kept separate, allowing different levels of disclosure without compromising the integrity of either.

Governance decisions are explainable without relying on model introspection — which matters, because introspection is unreliable.

4.6 Provide for or Cooperate in Remediation

Aurora-Lens supports deterministic replay of failure cases. Policy updates are versioned. Governance can be re-evaluated post-hoc against the original decision state, and the audit trace links the original governance action to any subsequent remediation.

Remediation is not an ad hoc process; it is structurally traceable.

5. Scope and Limitations

Aurora-Lens does not claim to constitute legal compliance certification. What it does is operationalise the procedural governance expectations defined in the OECD guidance — providing runtime enforcement and audit infrastructure to support RBC due diligence implementation.

It does not guarantee correctness of underlying model knowledge, replace organisational risk governance structures, or eliminate all forms of adversarial misuse. These are not gaps in the architecture; they reflect a deliberate boundary between what can be enforced mechanically and what requires organisational judgment.

6. Discussion

The OECD guidance defines governance as a process. Aurora-Lens defines governance as an enforceable mechanism. The distinction matters.

Process alone does not prevent runtime failure. A documented due diligence procedure can be followed correctly and still result in a confident, structurally unjustified model output being delivered to a user. Mechanism without governance context, on the other hand, lacks regulatory legitimacy — it is enforcement without meaning.

Aurora-Lens bridges this gap by translating procedural due diligence into deterministic runtime control. The transport-layer architecture makes this provider-agnostic and supports cross-jurisdictional policy adaptation without requiring changes to underlying models. In regulated environments, this directly reduces liability exposure from the specific class of failure the guidance is most concerned with: outputs that are confident, structurally unjustifiable, and already delivered.

7. Conclusion

The OECD *Due Diligence Guidance for Responsible AI* establishes a governance baseline for enterprises deploying AI systems. Aurora-Lens aligns with its six-step RBC framework by embedding governance into infrastructure, formalising risk detection, deterministically mitigating adverse impacts before delivery, maintaining tamper-evident audit records, supporting transparent communication of governance decisions, and enabling reproducible remediation.

By operating at the transport layer, Aurora-Lens operationalises procedural due diligence without modifying underlying models — making it applicable across providers and jurisdictions.

As AI governance converges toward enforceable accountability standards, the gap between policy-level guidance and runtime enforcement will need to close. Deterministic transport-layer governance is one viable path for closing it.